

in the consumer lighting industry are working to bring down the cost but, by the well-known rules of the market, prices can come down drastically only if there is mass-adoption. There sets in a vicious cycle!

Nevertheless, we do see quite a few connected, smart lighting consumer products popping up in international markets, and hope to see this segment grow in India too, in the near future. If you are wondering how a smart light could pep up your home, may be a description of some such products will give you a few nice ideas.

Philips Hue: Costly but exciting start to smart lighting

At close to \$200, the Philips Hue connected bulb starter kit does seem a bit overpriced. However, for that price you get three Zigbee based LED light bulbs and a Hue router, which mediates between the bulbs and your home Wi-Fi network. Philips' kit is positioned as a dynamic and capable system, which despite being very simple for beginners is also highly-programmable for those who wish to do more with it.

All you have to do is connect the Hue, fit the bulbs into the sockets and install an app on your mobile phone. The dots join by themselves, and you are ready to remotely control-and-schedule the colour, brightness and other features of your smart bulbs. With extended features like geo-fenc-



App with iLumi



ing and if-this-then-that (IFTTT), you can tune the system to your specific needs, like, for example, make the bulb in your kitchen blink when the oven timer strikes zero or set the study-room light to change colour when an email arrives in your inbox. If you are not much into programming, you can wait for somebody else to upload an app that matches your needs, because Philips has allowed third-parties to develop apps for Hue.

iLumi: An Indian mind at work

In June this year, iLumi Solutions received a patent from the United States Patent Office for their wireless lighting control system. Interestingly, iLumi, the award-winning, app-controlled LED bulb has an Indian mind behind it. In 2010, Swapnil Bora and Corey Egan, classmates at Naveen Jindal School of Management, University of Texas at Dallas, presented the concept at their college's annual Business Idea Competition. Their idea topped the charts and, with the prize money of \$5000,

iLumi was born!

The iLumi system is simpler than the Philips one because you do not need a hub with it. Just fit the bulb, install the app and you are ready to go. iLumi bulbs, which have a life of up to 20 years, offer bright and efficient multi-colour light with what the company calls HyperLux LED technology.

Each iLumi bulb has Bluetooth classic and Bluetooth low-energy (Bluetooth 4.0) built-in, along with a processor, memory and real-time clock.

With the mobile app, users can adjust, customise, program and schedule one or multiple iLumi bulbs. An iLumi bulb can recognise whether you are in the room or not and adjust lighting accordingly. It can also waver in sync with music or alert you of important information such as weather, news updates, stock price movements, phone calls or email messages!

LIFX: Colours life bright

Sometimes you just want something like, say, the good-old phone. Less features, but good at what it does! If that is what you expect of your smart bulb too, then LIFX is likely to be your choice.

The diodes in each LIFX bulb are capable of achieving a full spectrum of natural tones at up to 1000 lumens. '16 million colours at your fingertips,' is what the company claims. Generally, in coloured bulbs, the peak brightness is achieved by white, while the colours are much dimmer. With LIFX, you can add white light on top of the coloured light, which results in a full spectrum of tinted light to work with. LIFX bulbs consume 17 watts of energy and have a life of around 27 years. It is a bit heavier than its counterparts because it has a built-in heat sink.



LIFX bulb and phone